THE VISION

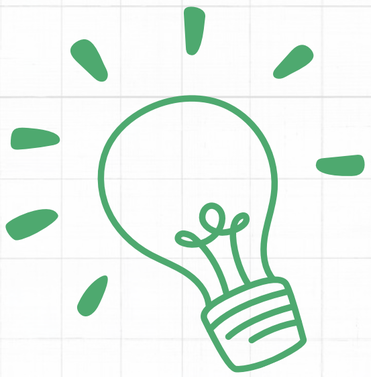
A

SPECTACLE

FOR

VISUALLY IMPAIRED

THE PROJECT

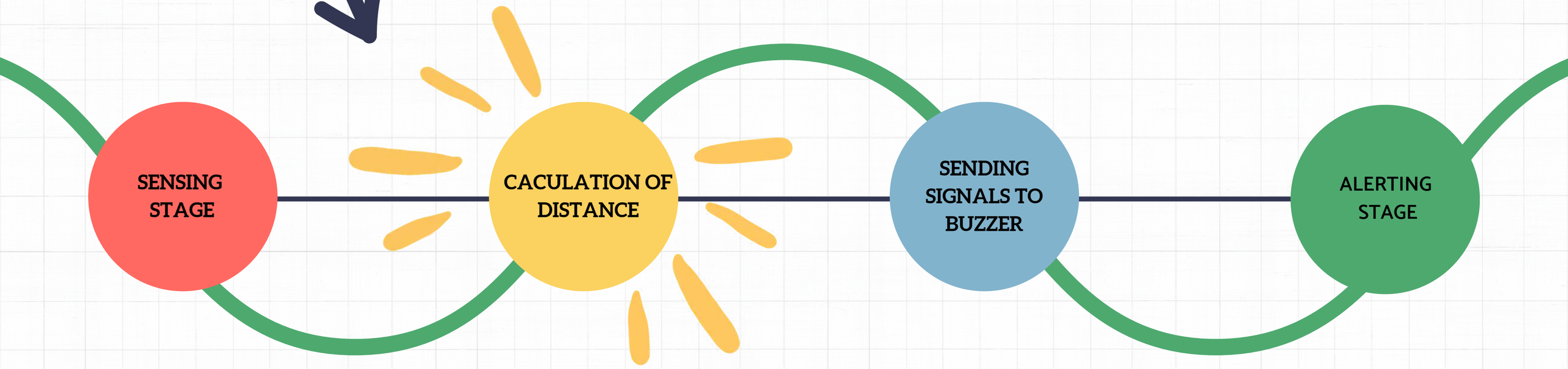


This project focuses on creating Ultrasonic Glasses for **visually impaired individuals**, which serve as a wearable mobility aid designed to alert the user about obstacles through sound patterns.

OBJECTIVES

- To design wearable ultrasonic glasses that detect obstacles in the front-left and front-right regions.
- To develop a program that controls buzzer delay based on real-time distance measurements
- To provide real-time audio alerts that vary based on obstacle distance.
- To ensure the system remains low-cost, lightweight, and power efficient.
- To synchronize both buzzers when an object is detected between both sensors (central position).
- To ensure efficient and stable operation powered by a single 9V battery.

METHODOLOGY

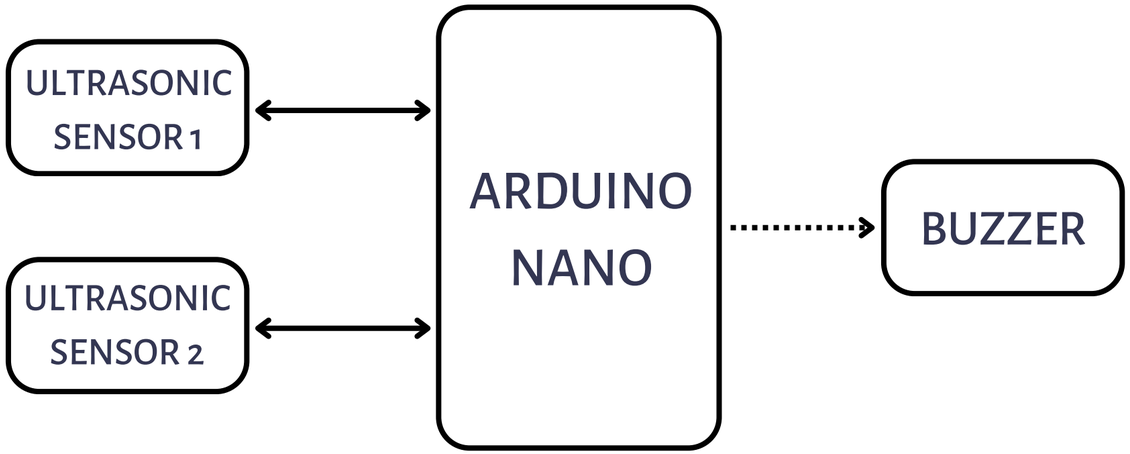


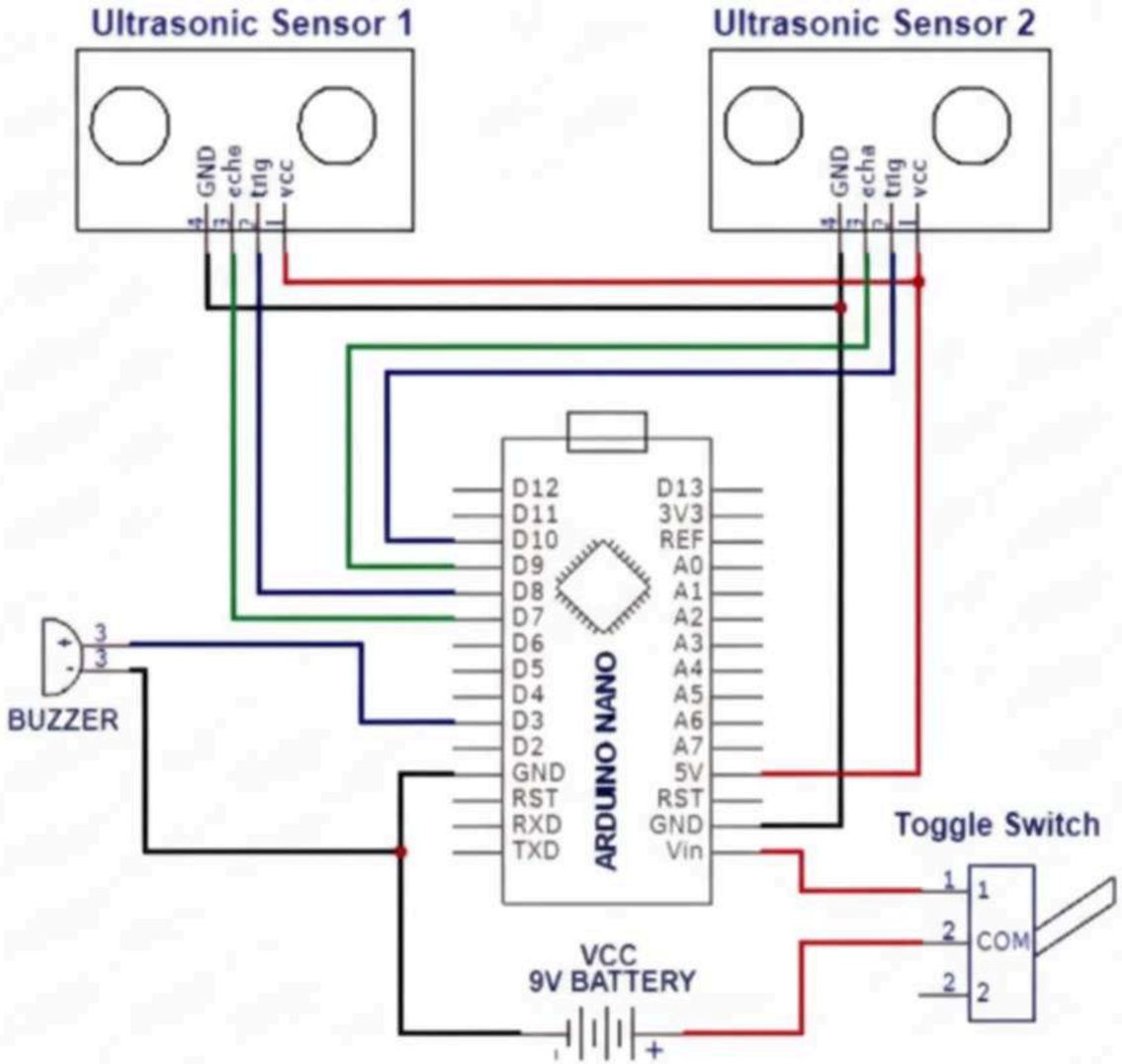
Ultrasonic Sensors

Arduino nano

Piezo Buzzer

CIRCUIT DIAGRAM





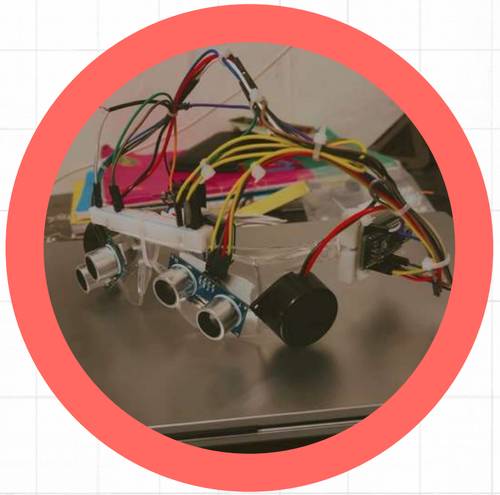
FEATURES

REAL-TIME OBSTACLE DETECTION

The glasses continuously scan the path ahead using sensors to identify obstacles in real time. This allows the user to receive instant warnings before coming into physical contact with objects.

HANDS-FREE WEARABLE DESIGN

All components are embedded into a lightweight spectacle frame. This keeps the user's hands free, making movement more natural and comfortable compared to carrying external devices.





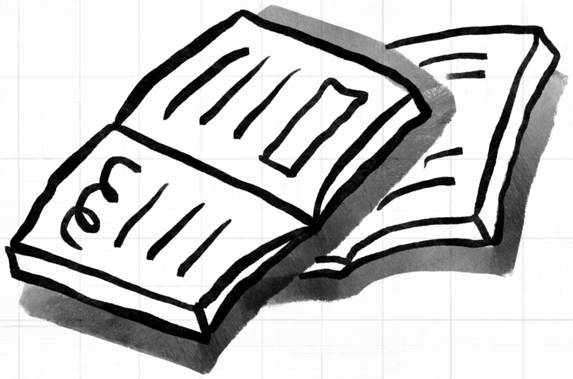
deepai.org

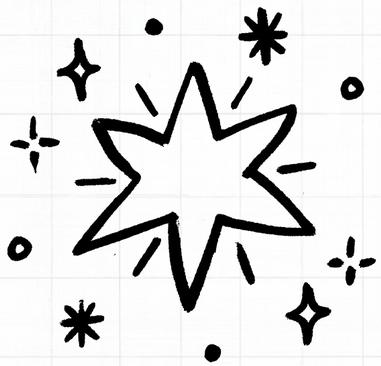
DISTANCE-BASED ALERTS

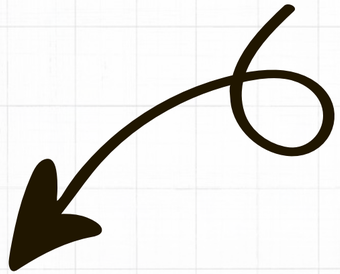
The glasses provide graded feedback depending on how close an obstacle is. As the object approaches, the warnings intensify – helping the user estimate distance without needing vision.

DIRECTIONAL OBSTACLE WARNING

The glasses can identify where the obstacle is located – on the left, right, or center and alert the user accordingly. By activating the buzzer only on the side where the obstacle is detected, the user can intuitively understand the direction and avoid it without confusion.







THANK YOU !!!

